Adjacent-charge propagation as Gaussian smoothing
A finite-time transfer estimate with all contractions retained

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Abstract

The ordinary charge-removal propagator admits an approximation by Gaussian Ornstein–Uhlenbeck smoothing of the propagator in the original charge sector. Its annealed normalized Hilbert–Schmidt error is $O(N^{-1})$ on compact real-time intervals, after the full infinite-temperature charge average. We first compute both directional transfer defects, including the nonzero reverse contraction source. Polar normalization then gives a small creation defect; adjoints and Gaussian-chaos tails extend it to the complete ordinary evolution. The preceding rank–chaos bound converts the resulting rank filter to a disorder-space smoothing. This is a propagation result; force insertions and their energy-regressed fluctuations remain the next calculation.

1 Model and principal statement

Use standard CAR, $\hbar = 1$, string time, and $H = \sum_{I,J} J_{IJ} c_I^\dagger c_J$, with overlapping p-body index sets, $J = J^\dagger$, and

$$\mathbb{E} J_{IJ} J_{KL} = w_N \delta_{IL} \delta_{JK}, \quad w_N = 4^p \frac{J_s^2}{\lambda_N} \left(\frac{p!}{N^p} \right)^2, \qquad \lambda_N = p^2/N. \tag{1}$$

All charge sectors share the same J. Write $h = H_m$, $h_- = H_{m-1}$, $\|X\|_{2,m}^2 = \mathbb{E}\,\mathrm{tr}_m\, X^\dagger X$, $\mathrm{tr}_m = \mathrm{Tr}\,/\binom{N}{m}$, and $b_m = 2^{-N}\binom{N}{m}$. Define

$$T_m(X) = \frac{1}{m} \sum_i c_i^\dagger X c_i, \qquad F_m(s) = T_m(e^{ish_-}), \qquad \alpha_m = (1 - p/m)_+. \tag{2}$$

For $0 \le a \le 1$, the disorder Ornstein–Uhlenbeck map is

$$(\mathcal{O}_a f)(J) = \mathbb{E}_{J'} f(aJ + \sqrt{1-a^2} J'), \qquad J' \text{ an independent copy.} \tag{3}$$

As $N \to \infty$, λ_N stays in a compact subset of $(0, \infty)$, and J_s and S are fixed. We prove

$$\boxed{\sup_{|s| \le S} \sum_{m=1}^{N} b_m \|F_m(s) - \mathcal{O}_{\alpha_m}(e^{ish})\|_{2,m}^2 = O_S(N^{-2}).} \tag{4}$$

The approximation preserves the original sample J; only the conditional average in (3) uses a new copy.

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2 The channel spectrum and a forward defect

For the finite identities assume $m > p \geq 1$. Set

$$n = N - m + 1, \quad \alpha = (m-p)/m, \quad C = w_N \binom{m}{p}\binom{N-m+p}{p}, \quad C_- = C\alpha\frac{n+p}{n}. \tag{5}$$

Subscripts $+$ and $-$ denote the upper and lower charge operator spaces with their normalized Hilbert–Schmidt inner products. The adjoint of $T = T_m$ is

$$A = T^\dagger, \qquad A(Y) = \frac{1}{n}\sum_i c_i Y c_i^\dagger. \tag{6}$$

Let $P_r^\pm$ be the multiplicity-free neutral $U(N)$ rank projectors, and $\mathcal{X}_\pm = \sum_r x_r P_r^\pm$, $x_r = r(N{+}1{-}r)$. Then

$$AT = 1 - \frac{\mathcal{X}_-}{mn}, \qquad TA = 1 - \frac{\mathcal{X}_+}{mn}, \qquad T\mathcal{X}_- = \mathcal{X}_+ T. \tag{7}$$

These spectra follow from the known embedded-GUE covariance spectrum [1]: the one-body case gives TA directly. CAR, or the primitive-rank norm ratio, gives AT. A rank present on only one side has $x_r = mn$ and is killed by the corresponding map. Both T and A are unital, trace preserving in these normalized traces, and contractions. The spectrum itself is an existing result's application.

Use an independent pair (u, v) sharing one new Gaussian coupling in the two charges. Gaussian covariance and CAR give the charged interface

$$Tv = \alpha u, \qquad \mathbb{E}(u c_i^\dagger v) = \alpha C c_i^\dagger, \qquad \mathbb{E}\, uT(vX) = \alpha C T X. \tag{8}$$

For deterministic lower-rank input X_r, the one-step Casimir mean and (7) imply

$$\mathbb{E}\|T(vX_r)\|_+^2 = \left(\alpha^2 C - \frac{\kappa_1^- x_r}{mn}\right)\|X_r\|_-^2, \qquad \kappa_1^- = \frac{C(m-p)(m-p-1)}{m(m-1)}. \tag{9}$$

Subtracting the two cross terms using (8) proves

$$\boxed{\mathbb{E}\|T(vX_r) - \alpha u T X_r\|_+^2 = \gamma x_r \|X_r\|_-^2, \qquad \gamma = \frac{Cp(m-p)}{m^3(m-1)n}.} \tag{10}$$

In real Gaussian coordinates $u = \sum_\beta g_\beta B_\beta^+$, $v = \sum_\beta g_\beta B_\beta^-$, with Hermitian coefficients and $\sum_\beta (B_\beta^\pm)^2 = C_\pm I$, this is an exact coefficient quadratic form. Its defining positive operator is equivariant, so different input ranks have zero cross terms. If X depends on the original J, the new copy is averaged with X held fixed.

3 The reverse source is explicit and orthogonal

The reverse-adjoint direction contains a source even on the identity. Put

$$V = v - \alpha A(u) = \frac{\mathcal{X}_- v}{mn}, \qquad D_u(Y) = A(uY) - A(u)A(Y). \tag{11}$$

There is an exact decomposition

$$vA(Y) - \alpha A(uY) = VA(Y) - \alpha D_u(Y). \tag{12}$$

The two terms are orthogonal in mean:

$$\mathbb{E}\langle VA(Y), D_u(Y)\rangle_- = 0. \tag{13}$$

Indeed the equivariant map $Z \mapsto \mathbb{E}[VZu]$ acts by one scalar j_V on the unique fundamental charged type c_i. Thus $\mathbb{E}[VA(uY)] = j_V A(Y)$ and $\mathbb{E}[VA(u)] = j_V I$, proving (13). No independence of the two terms is used.

For clarity the remaining coefficient can also be computed without new recoupling data. The upper one-step Casimir law has $\mu_1^+ = (m-p)(n-1)/[m(n+p-1)]$ and identity mean $c_+ = p(n-1)(N+1)/(n+p-1)$. Put

$$C_A = \mathbb{E}\,\mathrm{tr}_-[A(u)]^2 = C\left(1 - \frac{c_+}{mn}\right). \tag{14}$$

The fundamental charged covariance gives $\mathbb{E}[uc_i^\dagger A(u)] = C_A c_i^\dagger$. Consequently

$$\begin{aligned}\mathbb{E}\|D_u(Y_r)\|_-^2 &= \left\{C\left[1 - \frac{\mu_1^+ x_r + c_+}{mn}\right] - C_A\left[1 - \frac{x_r}{mn}\right]\right\}\|Y_r\|_+^2 \\ &= \frac{Cpx_r}{mn^2(n+p-1)}\|Y_r\|_+^2.\end{aligned} \tag{15}$$

The known second Casimir moment for v gives

$$\nu_0 = \mathbb{E}\,\mathrm{tr}_-\, V^2 = \frac{Cp^2(m-p)N(N+1)}{m^3 n(n+p-1)}. \tag{16}$$

Equations (12)–(13) therefore determine the entire reverse quadratic form:

$$\boxed{\mathbb{E}\|vA(Y_r) - \alpha A(uY_r)\|_-^2 = \left[\nu_0\left(1 - \frac{x_r}{mn}\right) + \frac{\alpha^2 Cpx_r}{mn^2(n+p-1)}\right]\|Y_r\|_+^2.} \tag{17}$$

At central filling the source has norm of order p/N, whereas the centered defect on ranks $r \le kp$ has squared norm $O(C\lambda_N k/N^2)$. Simply copying the forward estimate to annihilation would lose the source. Polar normalization handles both directions while retaining its effect in the channel amplitude.

4 Polar normalization makes the dynamical defect small

Take the polar decomposition on normalized operator Hilbert spaces,

$$T = \mathcal{U}S_- = S_+\mathcal{U}, \qquad S_\pm = \sqrt{1 - \mathcal{X}_\pm/(mn)}, \qquad s_r = \sqrt{\frac{(m-r)(n-r)}{mn}}. \tag{18}$$

$\mathcal{U}$ is a deterministic, rank-preserving partial isometry and $\mathcal{U}I = I$. We never invert s_r at a zero mode. In this and the next section take $N \geq 16$, $p \leq N/16$, $N/4 \leq m \leq 3N/4$, and

$$B_m = \lambda_N + \lambda_N^2 + \lambda_N \frac{(m-n)^2}{N}. \tag{19}$$

For $r \leq kp$ and $(k+1)p \leq N/8$, all ranks reached by one multiplication are common to both charges, and $s_t \geq 1/2$. Writing $E_\beta = TB_\beta^- - \alpha B_\beta^+ T$ as left multiplication maps, the rank-r to rank-t identity is

$$P_t^+(\mathcal{U}B_\beta^- - B_\beta^+\mathcal{U})X_r = \frac{P_t^+ E_\beta X_r + (\alpha s_r - s_t)P_t^+(B_\beta^+ \mathcal{U}X_r)}{s_t}. \tag{20}$$

The product triangle gives $\delta = p + r - t \in [0, 2p]$. The preceding rank-chain result gives, for the upper multiplication law,

$$\mathbb{E}_r \delta^2 \leq 10^6[\lambda_N(k+1) + \lambda_N^2(k+1)^2]. \tag{21}$$

On this interval $|s_t - s_{r+p}| \leq 32\delta/N$. Also, exactly,

$$\alpha^2 s_r^2 - s_{r+p}^2 = \frac{p}{mn}\left[\alpha(m-n) + \frac{2r(n-r)}{m} - \frac{px_r}{m^2}\right], \tag{22}$$

so that

$$|\alpha s_r - s_{r+p}| \leq \frac{32p|m-n|}{N^2} + \frac{640kp^2}{N^2}. \tag{23}$$

Combine these estimates, (10), and $\gamma x_r \leq 4096C\lambda_N k/N^2$ in (20). Different input-rank cross terms vanish by equivariance. For Gaussian creation $\mathcal{C}_\pm = \sum a_\beta^\dagger L_{B_\beta^\pm}$, symmetrization of a degree-k coefficient tensor costs only $k+1$:

$$\|(\mathcal{U}\mathcal{C}_- - \mathcal{C}_+\mathcal{U})\psi_k\|^2 \leq \frac{K_0 C B_m}{N^2}(k+1)^3\|\psi_k\|^2, \qquad K_0 = 10^{12}. \tag{24}$$

This constant is deliberately loose. The filling drift in B_m must be retained; a uniform $O(N^{-1})$ norm claim throughout the entire broad central interval would be incorrect.

5 All chaoses and the full real-time propagator

Work in the Gaussian L^2 space of equivariant matrix functions; $\mathcal{N}$ is the chaos number. Degree k has matrix rank at most kp. Put $K = \lfloor N/(8p) \rfloor - 1$. On degrees at most $K+1$, the partial isometry is unitary between the complete equivariant subspaces. If $E = \mathcal{U}\mathcal{C}_- - \mathcal{C}_+\mathcal{U}$, the annihilation defect on these blocks is

$$\mathcal{U}\mathcal{C}_-^\dagger - \mathcal{C}_+^\dagger\mathcal{U} = \mathcal{U}E^\dagger\mathcal{U}. \tag{25}$$

Thus (24) controls both directions. Gaussian creation and annihilation conventions are standard [2]. Let $\mathcal{H}_\pm = L_{H_\pm} = \mathcal{C}_\pm + \mathcal{C}_\pm^\dagger$ and $G = \mathcal{U}\mathcal{H}_- - \mathcal{H}_+\mathcal{U}$. With $M = \max_j C_j$,

$$\|G\psi_{\le K}\|^2 \le \frac{4K_0MB_m}{N^2}\|(\mathcal{N}+1)^{3/2}\psi_{\le K}\|^2, \tag{26}$$

$$\|G\psi_{>K}\|^2 \le 16M\|(\mathcal{N}+1)^{1/2}\psi_{>K}\|^2. \tag{27}$$

The second line uses only $\|\mathcal{U}\| \le 1$ and $\|\mathcal{H}_\pm\psi\|^2 \le 4C_\pm\|(\mathcal{N}+1)^{1/2}\psi\|^2$.

For $\psi(t) = e^{ith_-}$, the preceding ordinary-propagator proof gives

$$\sum_j \zeta^j\|\psi_j(t)\|^2 \le e^{C_-t^2(\zeta-1)}, \quad \zeta \ge 1, \tag{28}$$

$$\|(\mathcal{N}+1)^{3/2}\psi(t)\|^2 \le P_3(C_-t^2), \qquad P_3(z) = 1 + 7z + 6z^2 + z^3. \tag{29}$$

Its factorial moments also give $\|(\mathcal{N}+1)^{1/2}\psi_{>K}\|^2 \le 2^{-(K+1)}(1+2z)e^z$, $z = C_-t^2$. For $|s| \le S$, define $z_S = MS^2$ and

$$R_m(S) = S^2\left[\frac{8K_0MB_m}{N^2}P_3(z_S) + 32M2^{-(K+1)}(1+2z_S)e^{z_S}\right]. \tag{30}$$

The exact Hilbert-space Duhamel formula, using $\mathcal{U}I = I$, is

$$\mathcal{U}e^{ish_-} - e^{ish} = i\int_0^s e^{i(s-t)h}G(e^{ith_-})\,dt. \tag{31}$$

Finite-N Gaussian Sobolev domains justify the formula; the left factor is unitary for real time. Hence

$$\|\mathcal{U}e^{ish_-} - e^{ish}\|_+^2 \le R_m(S), \qquad \|F_m(s) - S_+e^{ish}\|_+^2 \le R_m(S). \tag{32}$$

For fixed $\lambda_N > 0$ the chaos-tail term is exponentially small in $\sqrt{N}$. No truncated time series is used.

6 From the physical-rank filter to disorder OU smoothing

This step uses the full ordinary rank–chaos theorem from the preceding stage. Define $\mathcal{R} = \sum_r r P_r^+$ and $\mathfrak{D} = p\mathcal{N} - \mathcal{R} \geq 0$ on equivariant functions. For every allowed upper rank, $u = x_r/(mn) \in [0,1]$, and

$$0 \leq 1 - u/2 - \sqrt{1-u} \leq u^2/2. \tag{33}$$

Using $m, n \geq N/4$, $x_r = r(m+n-r)$, and $N+1 \leq 2N$, one obtains

$$|s_r - (1 - r/m)| \leq \frac{8r|m-n|}{N^2} + \frac{520r^2}{N^2}. \tag{34}$$

For integer $k \geq 0$, Taylor's theorem on $0 \leq p/m \leq 1$ gives $|\alpha^k - 1 + kp/m| \leq k(k-1)p^2/(2m^2)$ (zero for $k = 0, 1$). Consequently, for all $r \leq kp$, without a low-rank restriction,

$$\boxed{|s_r - \alpha^k| \leq \frac{4(kp-r)}{N} + \frac{8kp|m-n|}{N^2} + \frac{528k^2p^2}{N^2}.} \tag{35}$$

Orthogonal rank and chaos decompositions then give

$$\begin{aligned} \|(S_+ - \alpha^{\mathcal{N}})e^{ish}\|_+ \leq \frac{4}{N}\|\mathfrak{D}e^{ish}\|_+ + \frac{8p|m-n|}{N^2}\|\mathcal{N}e^{ish}\|_+ \\ + \frac{528p^2}{N^2}\|\mathcal{N}^2 e^{ish}\|_+. \end{aligned} \tag{36}$$

For $z = Cs^2$, the established bounds are

$$\|\mathfrak{D}e^{ish}\|_+^2 \leq 4 \cdot 10^6 C(\lambda_N + \lambda_N^2)s^2 P_3(z), \tag{37}$$

$$\|\mathcal{N}e^{ish}\|_+^2 \leq z + z^2, \tag{38}$$

$$\|\mathcal{N}^2 e^{ish}\|_+^2 \leq P_4(z), \qquad P_4(z) = z + 7z^2 + 6z^3 + z^4. \tag{39}$$

Thus a finite bound for the squared filter error is

$$\begin{aligned} Q_m(S) = \frac{192 \cdot 10^6 M(\lambda_N + \lambda_N^2)S^2}{N^2} P_3(z_S) \\ + \frac{192\lambda_N(m-n)^2}{N^3}(z_S + z_S^2) + \frac{836352\lambda_N^2}{N^2} P_4(z_S). \end{aligned} \tag{40}$$

Since $\alpha^{\mathcal{N}} = \mathcal{O}_\alpha$, combining with (32) yields the central-sector bound

$$\|F_m(s) - \mathcal{O}_\alpha(e^{ish})\|_+^2 \leq 2R_m(S) + 2Q_m(S). \tag{41}$$

This smoothing acts on the Gaussian coupling, not on the spectral energy variable. Its parameter α is distinct from the previous energy-Mehler parameter $q = e^{-4\lambda}$.

7 Full charge average and the next memory calculation

The constants remain uniform in the full-charge ensemble. Indeed

$$C_m \le \frac{J_s^2}{\lambda_N} \prod_{j=0}^{p-1} \left(1 + \frac{p-2j}{N}\right)^2 \le \frac{J_s^2}{\lambda_N} e^{2p/N}. \tag{42}$$

The first inequality uses $4ab \le (a+b)^2$ on each positive factor in the two binomials; sectors with $m < p$ have $C_m = 0$. Under b_m, $\mathbb{E}_b(m-n)^2 = N+1$, so

$$\sum_{m \text{ central}} b_m B_m \le \lambda_N + \lambda_N^2 + \lambda_N(1 + 1/N). \tag{43}$$

Outside $[N/4, 3N/4]$ the total binomial weight is at most $2e^{-N/8}$. At $0 < m \le p$, $F_m = I$; at $m = p$ the OU proxy need not equal I. The true F_m has normalized HS norm at most one. The OU proxy also has annealed norm at most one. Hence the exterior contribution to the squared error is at most $8e^{-N/8}$. Use the clipped α_m there. Equation (41) now proves (4). The absent charge-zero removal term may equivalently be assigned zero error.

The new interface. The approximation replaces a neighboring-charge exponential by an explicit conditional Gaussian average in the original sector:

$$\mathcal{O}_\alpha(e^{ish(J)}) = \mathbb{E}_{J'} e^{is[\alpha h(J) + \sqrt{1-\alpha^2} h(J')]}. \tag{44}$$

It retains every coupling chaos and the original common disorder. It does not assert samplewise commutation with $h(J)$.

For the actual force, $a = \alpha h$ and

$$\Lambda(s) = -F_m''(s) + i\{a, F_m'(s)\} + aF_m(s)a. \tag{45}$$

Differentiating the *OU proxy itself* is exact and gives, with $H' = \alpha h + \sqrt{1-\alpha^2}\, u$,

$$\Lambda_{\mathrm{OU}}(s) = (1-\alpha^2)\mathbb{E}_u\big[u e^{isH'} u\big]. \tag{46}$$

The independent u in (46) is inside the same exponential as its two insertions; it cannot be replaced by an independent covariance sandwich. This is an explicit next target.

The proved error is currently unweighted L^2. It cannot be differentiated to estimate $\Lambda - \Lambda_{\mathrm{OU}}$, nor multiplied freely by h. Moreover, the collective factor m makes an $O(N^{-1})$ force error order one. Next we must obtain the required derivative/energy-weighted transfer bounds and identify the corresponding fluctuation, before discarding any such term. The finite-time X_0 covariance and bulk meson dictionary remain open.

8 Evidence and literature boundary

The spectrum in (7) is the one-body specialization of the established embedded-GUE spectrum, not a new spectrum. The Gaussian creation, adjoint and OU facts are standard. This stage applies them to the concrete common-coupling adjacent-charge map: its exact directional defects, the low-degree polar estimate, the complete real-time propagation bound and the OU comparison. The proofs are written above; finite examples do not establish their uniform asymptotic scope.

Saved verification. Six systems $(N, p, m) = (3, 1, 2), (4, 1, 2), (4, 2, 3), (5, 1, 2), (5, 2, 3), (5, 2, 4)$ were constructed directly from CAR matrices. For each, every real Gaussian coordinate and the full matrix-input space were retained. Complete superoperator checks verify both channel Grams, the forward defect, both polar maps, the centered reverse defect, its full reverse quadratic form and the complex source–defect orthogonality. These are not Monte Carlo checks or fits of a large-N rate. Two rational identities check the forward coefficient, one polynomial identity checks the polar amplitudes, and 230 exact rational cases check the source variance from the known Casimir second moment.

The three scientific JSON files are saved in `results/susskind-memory-charged-transfer/`. Their drivers record parameter sets, dependency versions and input hashes and refuse to overwrite existing outputs. Small systems also include unequal operator-space dimensions and channel zero modes. The previous ordinary rank–chaos evidence and all older outputs remain unchanged. Final source, PDF, two-pass build log and page renders are bound by a separate local evidence record.

Sources and next checks. Kota's covariance result is read at finite N, without importing a dilute-filling approximation. Chen–Liu is used for the explicit Gaussian adjoint convention and Hermite structure. Standard CP Schwarz structure was also checked against Choi's original paper; the new rate is not obtained from a generic CP contraction alone. The complex-SYK finite-site rules remain the complementary route to a direct force four-point function. Their leading neutral-probe formula cannot replace the charged, subleading calculation.

References

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